Alexander B. Thames, PhD

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Summary

Geoscientist and climate scientist with doctoral experience in developing tools and methodologies to support assessments of deep uncertainty, particularly for geophysical models and climate change impact analyses on natural-human systems. Expertise in large ensemble exploratory modeling in the context of understanding novel system interactions, informing decision-making and risk, high-performance computing, and experimental atmospheric chemistry and physics.

Education

PhD The Pennsylvania State University, Geosciences w/ Climate Science Dual-Title — Aug 2019 – May 2026

- *Dissertation*: Exploratory Modeling: An Interdisciplinary Tool for Tackling Deep Uncertainty Across Earth System Science

BSc The Pennsylvania State University, Physics w/ Mathematics Minor — Aug 2011 – May 2015

Experience

Research Scientist I
Postdoctoral Associate
CIRES, Chemical Sciences Laboratory
Nov 2026 - present
Jun 2026 - Oct 2026
Atmospheric Composition & Chemical Processes

- Supported the Single Particle Soot Photometer instrument with calibrations, fieldwork, and data processing for the following NASA- and NOAA-funded airborne research campaigns:
  - INjected Smoke and PYRocumulonimbus Experiment (INSPYRE)
  - Stratospheric Aerosol processes, Budget and Radiative Effects (SABRE)

PhD Research
Aug 2019 - May 2026
Dr. Bradford Foley and Dr. Antonia Hadjimichael Research Groups

- Developed a stochastic weather generator in Python for statistically generating new realizations of precipitation and temperature from historical observations and regional CMIP6 downscaled projections
- Integrated weather generator into a regional hydrologic/agricultural consumptive use model to explore plausible agricultural risks and sensitivities of users and crops to climate in an institutionally complex river basin in the American west
- Constructed an exploratory geophysical model in MATLAB for investigating how Earth's internal and external water reservoirs affect its thermal history and confine the Earth's total water budget through the non-classical feedback of variable regassing efficiency
- Optimized, parallelized, validated, and visualized experiment design for high dimensional climate models to improve and inform agricultural decision-making, and geophysical models to understand how new feedbacks in coupled mantle temperature-water models impact Earth's thermal evolution

Associate Research Scientist
The Pennsylvania State University, collaborating with NASA, NOAA CSL, & more
Jun 2015 - Aug 2019
Dr. William Brune Research Group

- Led fieldwork as primary operator of Airborne Tropospheric Hydrogen Oxides Sensor and OH reactivity instrument across one international intercomparison study (SAPHIR OHR Intercomparison) and two NASA-funded airborne research campaigns, Korea-US Air Quality (KORUS-AQ) and Atmospheric Tomography (ATom)
- Collected, processed, and analyzed *in situ* measurements using bespoke MATLAB

software for over 600+ flight-research hours across all seven continents

Skills

Python, MATLAB, Linux/Unix, HPC (SLURM), GDScript, GLSL, C#, Igor, SQL	Programming
LaTeX, Git, Illustrator, Markdown, Microsoft Office Suite, Blender, QGIS	Software and Tools
English (fluent), Spanish (A2)	Languages

Publications

Published Manuscripts, First Author

Thames, A.B., Hadjimichael, A., & Quinn, J.D. (2026). Climate sensitivity of agricultural water demand depends on control over growing season: Implications for producers in the Upper Colorado River Basin. *Water Resources Research*, 62, e2026WR043402. 10.1029/2026WR043402 — 2026

Thames, A.B., & Foley, B.J. Reconciling Coupled Thermal-Water Evolution Models of Earth with Observations through Variable Regassing Efficiency. *Journal of Geophysical Research: Solid Earth*, 131, e2025JB033701. 10.1029/2025JB033701 — 2026

Thames, A.B., Hadjimichael, A., and Quinn, J.D.: `swxg`: A Python Library for Generalized Multivariate, Multisite, Copula-Based Stochastic Weather Generation. *Journal of Open Research Software*, 14(1), 47. 10.5334/jors.651 — 2026

Thames, A.B., Brune, W.H., Miller, D.O., Allen, H.M., Apel, E.C., Blake, D.R., Bui, T.P., Commane, R., Crounse, J.D., Daube, B.C., Diskin, G.S., DiGangi, J.P., Elkins, J.W., Hall, S.R., Hanisco, T.F., Hannun, R.A., Hintsa, E., Hornbrook, R.S., Kim, M.J., McKain, K., Moore, F.L., Nicely, J.M., Peischl, J., Ryerson, T.B., St. Clair, J.M., Sweeney, C., Teng, A., Thompson, C.R., Ullmann, K., Wennberg, P.O., and Wolfe, G.M.: Missing OH reactivity in the global marine boundary layer. *Atmospheric Chemistry and Physics*, 20(6), 4013-4029. 10.5194/acp-20-4013-2020 — 2020

Published Manuscripts, Contributing Author

Baublitz, C.B., Fiore, A.M., Ludwig, S.M., Nicely, J.M., Wolfe, G.M., Murray, L.T., Commane, R., Prather, M.J., Anderson, D.C., Correa, G., Duncan, B.N., Follette-Cook, M., Westervelt, D.M., Bourgeois, I., Brune, W.H., Bui, T.P., DiGangi, J.P., Diskin, G.S., Hall, S.R., McKain, K., Miller, D.O., Peischl, J., **Thames, A.B.**, Thompson, C.R., Ullmann, K., and Wofsy, S.C.: An observation-based, reduced-form model for oxidation in the remote marine troposphere. *Proceedings of the National Academy of Sciences*, 120(34), e2209735120. 10.1073/pnas.2209735120 — 2023

Thompson, C.R., Wofsy, S.C., Prather, M.J., Newman, P.A., Hanisco, T.F., Ryerson, T.B., Fahey, D.W., Apel, E.C., Brock, C.A., Brune, W.H., Froyd, K., Katich, J.M., Nicely, J.M., Peischl, J., Ray, E., Veres, P.R., Wang, S., Allen, H.M., Asher, E., Bian, H., Blake, D., Bourgeois, I., Budney, J., Bui, T.P., Butler, A., Campuzano-Jost, P., Chang, C., Chin, M., Commane, R., Correa, G., Crounse, J.D., Daube, B., Dibb, J.E., DiGangi, J.P., Diskin, G.S., Dollner, M., Elkins, J.W., Fiore, A.M., Flynn, C.M., Guo, H., Hall, S.R., Hannun, R.A., Hills, A., Hintsa, E.J., Hodzic, A., Hornbrook, R.S., Huey, L.G., Jimenez, J.L., Keeling, R.F., Kim, M.J., Kupc, A., Lacey, F., Lait, L.R., Lamarque, J.-F., Liu, J., McKain, K., Meinardi, S., Miller, D.O., Montzka, S.A., Moore, F.L., Morgan, E.J., Murphy, D.M., Murray, L.T., Nault, B.A., Neuman, J.A., Nguyen, L., Gonzalez, Y., Rollins, A.W., Rosenlof, K., Sargent, M., Schill, G.P., Schwarz, J.P., St Clair, J.M., Steenrod, S.D., Stephens, B.B., Strahan, S.E., Strode, S.A., Sweeney, C., **Thames, A.B.**, Ullmann, K., Wagner, N., Weber, R., Weinzierl, B., Wennberg, P.O., Williamson, C.J., Wolfe, G.M., and Zeng, L.: The NASA Atmospheric Tomography (ATom) mission: Imaging the chemistry of the global atmosphere. *Bulletin of the American Meteorological Society*, 103(3), E761-E790. — 2022

10.1175/BAMS-D-20-0315.1

Brune, W.H., Miller, D.O., **Thames, A.B.**, Brosius, A.L., Barletta, B., Blake, D.R., Blake, N.J., Chen, G., Choi, Y., Crawford, J.H., DiGangi, J.P., Diskin, G., Fried, A., Hall, S.R., Hanisco, T.F., Huey, G.L., Hughes, S.C., Kim, M., Meinardi, S., Montzka, D.D., Pusede, S.E., Schroeder, J.R., Teng, A., Tanner, D.J., Ullmann, K., Walega, J., Weinheimer, A., Wisthaler, A., and Wennberg, P.O.: Observations of atmospheric oxidation and ozone production in South Korea. *Atmospheric Environment*, 269, 118854. 10.1016/j.atmosenv.2021.118854 — 2022

Kim, S., Seco, R., Gu, D., Sanchez, D., Jeong, D., Guenther, A.B., Lee, Y., Mak, J.E., Su, L., Kim, D.B., Lee, Y., Ahn, J.-Y., Mcgee, T., Sullivan, J., Long, R., Brune, W.H., **Thames, A.B.**, Wisthaler, A., Müller, M., Mikoviny, T., Weinheimer, A., Yang, M., Woo, J.-H., Kim, S., and Park, H.: The role of a suburban forest in controlling vertical trace gas and OH reactivity distributions–a case study for the Seoul metropolitan area. *Faraday discussions*, 226, 537-550. 10.1039/D0FD00081G — 2021

Kupc, A., Williamson, C.J., Hodshire, A.L., Kazil, J., Ray, E., Bui, T.P., Dollner, M., Froyd, K.D., McKain, K., Rollins, A., Schill, G.P., **Thames, A.B.**, Weinzierl, B.B., Pierce, J.R., and Brock, C.A.: The potential role of organics in new particle formation and initial growth in the remote tropical upper troposphere. *Atmospheric Chemistry and Physics*, 20(23), 15037-15060. 10.5194/acp-20-15037-2020 — 2020

Wang, S., Apel, E.C., Schwantes, R.H., Bates, K.H., Jacob, D.J., Fischer, E.V., Hornbrook, R.S., Hills, A.J., Emmons, L.K., Pan, L.L., Honomichl, S., Tilmes, S., Lamarque, J.-F., Yang, M., Marandino, C.A., Saltzman, E.S., de Bruyn, W., Kameyama, S., Tanimoto, H., Omori, Y., Hall, S.R., Ullmann, K., Ryerson, T.B., Thompson, C.R., Peischl, J., Daube, B.C., Commane, R., McKain, K., Sweeney, C., **Thames, A.B.**, Miller, D.O., Brune, W.H., Diskin, G.S., DiGangi, J.P., and Wofsy, S.C.: Global atmospheric budget of acetone: Air-sea exchange and the contribution to hydroxyl radicals. *Journal of Geophysical Research: Atmospheres*, 125(15), e2020JD032553. 10.1029/2020JD032553 — 2020

Travis, K.R., Heald, C.L., Allen, H.M., Apel, E.C., Arnold, S.R., Blake, D.R., Brune, W.H., Chen, X., Commane, R., Crounse, J.D., Daube, B.C., Diskin, G.S., Elkins, J.W., Evans, M.J., Hall, S.R., Hintsa, E.J., Hornbrook, R.S., Kasibhatla, P.S., Kim, M.J., Luo, G., McKain, K., Millet, D.B., Moore, F.L., Peischl, J., Ryerson, T.B., Sherwen, T., **Thames, A.B.**, Ullmann, K., Wang, X., Wennberg, P.O., Wolfe, G.M., and Yu, F.: Constraining remote oxidation capacity with ATom observations. *Atmospheric Chemistry and Physics*, 20(13), 7753-7781. 10.5194/acp-20-7753-2020 — 2020

Veres, P.R., Neuman, J.A., Bertram, T.H., Assaf, E., Wolfe, G.M., Williamson, C.J., Weinzierl, B., Tilmes, S., Thompson, C.R., **Thames, A.B.**, Schroder, J.C., Saiz-Lopez, A., Rollins, A.W., Roberts, J.M., Price, D., Peischl, J., Nault, B.A., Møller, K.H., Miller, D.O., Meinardi, S., Li, Q., Lamarque, J.-F., Kupc, A., Kjaergaard, H.G., Kinnison, D., Jimenez, J.L., Jernigan, C.M., Hornbrook, R.S., Hills, A., Dollner, M., Day, D.A., Cuevas, C.A., Campuzano-Jost, P., Burkholder, J., Bui, T.P., Brune, W.H., Brown, S.S., Brock, C.A., Bourgeois, I., Blake, D.R., Apel, E.C., and Ryerson, T.B.: Global airborne sampling reveals a previously unobserved dimethyl sulfide oxidation mechanism in the marine atmosphere. *Proceedings of the National Academy of Sciences*, 117(9), 4505-4510. 10.1073/pnas.1919344117 — 2020

Brune, W.H., Miller, D.O., **Thames, A.B.**, Allen, H.M., Apel, E.C., Blake, D.R., Bui, T.P., Commane, R., Crounse, J.D., Daube, B.C., DiGangi, J.P., Diskin, G.S., Elkins, J.W., Hall, S.R., Hanisco, T.F., Hannun, R.A., Hintsa, E.J., Hornbrook, R.S., Kim, M.J., McKain, K., Moore, F.L., Neuman, J.A., Nicely, J.M., Peischl, J., Ryerson, T.B., St. Clair, J.M., Sweeney, C., Teng, A.P., Thompson, C., Ullmann, K., Veres, P.R., Wennberg, P.O., and Wolfe, G.M. Exploring oxidation in the remote free troposphere: Insights from Atmospheric Tomog- — 2020

raphy (ATom). *Journal of Geophysical Research: Atmospheres*, 125(1), e2019JD031685. 10.1029/2019JD031685

Wolfe, G.M., Abad, G.G., Brune, W.B., Bui, P., Chang, C., Crounse, J.D., Dean-Day, J., Diskin, G., Hall, S.R., Hanisco, T.F., Kim, M., Liao, J., McKain, K., Miller, D., Nicely, J.M., Oman, L.D., Peischl, J., Ryerson, T.B., Sweeney, C., St. Clair, J.M., **Thames, A.B**, Thompson, C.R., Ullmann, K., and Wennberg, P.O.: Mapping hydroxyl variability throughout the global remote troposphere via synthesis of airborne and satellite formaldehyde observations. *Proceedings of the National Academy of Sciences*, 116(23), 11171-11180. 10.1073/pnas.1821661116 — 2019

Romer P.S., Wooldridge P.J., Crounse J.D., Kim M.J., Wennberg P.O., Dibb J.E., Scheuer E., Blake D.R., Meinardi S., Brosius A.L., **Thames A.B.**, Miller D.O., Brune W.H., Hall S.R., Ryerson T.B., and Cohen R.C.: Constraints on Aerosol Nitrate Photolysis as a Potential Source of HONO and NO_x. *Environ Sci Technol*, 52(23):13738-13746. 10.1021/acs.est.8b03861 — 2018

Fuchs, H., Novelli, A., Rolletter, M., Hofzumahaus, A., Pfannerstill, E.Y., Kessel, S., Edtbauer, A., Williams, J., Michoud, V., Dusanter, S., Locoge, N., Zannoni, N., Gros, V., Truong, F., Sarda-Esteve, R., Cryer, D.R., Brumby, C.A., Whalley, L.K., Stone, D., Seakins, P.W., Heard, D.E., Schoemaecker, C., Blocquet, M., Coudert, S., Batut, S., Fittschen, C., **Thames, A.B.**, Brune, W.H., Ernest, C., Harder, H., Muller, J.B. A., Elste, T., Kubistin, D., Andres, S., Bohn, B., Hohaus, T., Holland, F., Li, X., Rohrer, F., Kiendler-Scharr, A., Tillmann, R., Wegener, R., Yu, Z., Zou, Q., and Wahner, A.: Comparison of OH reactivity measurements in the atmospheric simulation chamber SAPHIR. *Atmos. Meas. Tech.*, 10, 4023–4053. 10.5194/amt-10-4023-2017 — 2017

Datasets

Thames, A.B. Output Data for Thames et al. – Climate Sensitivity of Agricultural Water Demand (1.0.1) [Dataset]. Zenodo. 10.5281/zenodo.18148466 — 2026

Thames, A.B. Input Data for Thames et al. – Climate Sensitivity of Agricultural Water Demand (1.0.0) [Dataset]. Zenodo. 10.5281/zenodo.18071209 — 2025

Thames, A.B.. Output Data for Thames & Foley – Reconciling Coupled Thermal-Water Evolution Models (1.0.0) [Dataset]. Zenodo. 10.5281/zenodo.17903920 — 2025

Brune, W.H., Miller, D.O., and **Thames, A.B.**: ATom: L2 Measurements from Airborne Tropospheric Hydrogen Oxides Sensor (ATHOS). ORNL DAAC, Oak Ridge, Tennessee, USA. 10.3334/ORNLDAAC/1709 — 2019

Wolfe, G.M., Abad, G.G., Brune, W.H., Bui, T.P., Chang, C.S., Crounse, J.D., Dean-Day, J.M., Diskin, G.S., Hall, S.R., Hanisco, T.F., Kim, M.J., Liao, J., Mccain, K., Miller, D.O., Nicely, J.M., Oman, L., Peischl, J., Ryerson, T.B., Sweeney, C., St.Clair, J.M., **Thames, A.B.**, Ullmann, K., Wennberg, P.: ATom: Column-Integrated Densities of Hydroxyl and Formaldehyde in Remote Troposphere. ORNL DAAC, Oak Ridge, Tennessee, USA. 10.3334/ORNLDAAC/1669 — 2019

Wofsy, S.C., Afshar, S., Allen, H.M., Apel, E.C., Asher, E.C., Barletta, B., Bent, J., Bian, H., Biggs, B.C., Blake, D.R., Blake, N., Bourgeois, I., Brock, C.A., Brune, W.H., Budney, J.W., Bui, T.P., Butler, A., Campuzano-Jost, P., Chang, C.S., Chin, M., Commane, R., Correa, G., Crounse, J.D., Cullis, P.D., Daube, B.C., Day, D.A., Dean-Day, J.M., Dibb, J.E., DiGangi, J.P., Diskin, G.S., Dollner, M., Elkins, J.W., Erdesz, F., Fiore, A.M., Flynn, C.M., Froyd, K.D., Gesler, D.W., Hall, S.R., Hanisco, T.F., Hannun, R.A., Hills, A.J., Hintsa, E.J., Hoffman, A., Hornbrook, R.S., Huey, L.G., Hughes, S., Jimenez, J.L., Johnson, B.J., Katich, J.M., Keeling, R.F., Kim, M.J., Kupc, A., Lait, L.R., Lamarque, J.-F., Liu, J., McKain, K., Mclaughlin, — 2018

R.J., Meinardi, S., Miller, D.O., Montzka, S.A., Moore, F.L., Morgan, E.J., Murphy, D.M., Murray, L.T., Nault, B.A., Neuman, J.A., Newman, P.A., Nicely, J.M., Pan, X., Paplawsky, W., Peischl, J., Prather, M.J., Price, D.J., Ray, E.A., Reeves, J.M., Richardson, M., Rollins, A.W., Rosenlof, K.H., Ryerson, T.B., Scheuer, E., Schill, G.P., Schroder, J.C., Schwarz, J.P., St.Clair, J.M., Steenrod, S.D., Stephens, B.B., Strode, S.A., Sweeney, C., Tanner, D., Teng, A.P., **Thames, A.B.**, Thompson, C.R., Ullmann, K., Veres, P.R., Vieznor, N., Wagner, N.L., Watt, A., Weber, R., Weinzierl, B., Wennberg, P.O., Williamson, C.J., Wilson, J.C., Wolfe, G.M., Woods, C.T., Zeng, L.H.: ATom: Merged Atmospheric Chemistry, Trace Gases, and Aerosols. ORNL DAAC, Oak Ridge, Tennessee, USA.
10.3334/ORNLDAAC/1581

Software

Thames, A.B. swxg: A Python Library for Generalized Multivariate, Multisite, Copula-Based Stochastic Weather Generation (0.4.0). Zenodo. 2025
10.5281/zenodo.17592197

Grants, Awards, and Certificates

Paul D. Krynine Scholarship 2020 - 2022, 2024 - 2025
The Pennsylvania State University

2nd Place, Oral Presentations 2025
Penn State Interdisciplinary Environmental Research Symposium

Alley Family Graduate Scholarship 2024
The Pennsylvania State University

NASA ARSET: Drought Monitoring, Prediction, and Projection 2024
National Aeronautics and Space Administration

Pottorf Endowment for Graduate Excellence 2024
The Pennsylvania State University

Michael Loudin Family Graduate Scholarship 2023
The Pennsylvania State University

Earle S. Lenker Award 2022 - 2023
The Pennsylvania State University

NASA Group Achievement Award | Atmospheric Tomography (ATom) 2019
The Pennsylvania State University

NASA Group Achievement Award | Korea-US Air Quality Mission (KORUS-AQ) 2016
The Pennsylvania State University

Presentations and Conferences

Thames, A.B., Hadjimichael, A., and Quinn, J.D.: Climate Sensitivity of Agricultural Water Demand Depends on Control Over Growing Season. *European Geosciences Union*, PICO Presentation. Online 2026

Thames, A.B., Hadjimichael, A., and Quinn, J.D.: Climate Sensitivity of Agricultural Water Demand Depends on Control Over Growing Season. *American Geophysical Union*, Poster Presentation. New Orleans, LA 2025

Thames, A.B., Hadjimichael, A., and Quinn, J.D.: Assessing Compound Climate Impacts to Agriculture in the Upper Colorado River Basin Using a Multisite Weather Generator. *Penn State Climate Solutions Symposium*, Poster Presentation. State College, PA 2025

Thames, A.B. and Hadjimichael, A.: Assessing Compound Climate Impacts on Agricultural Water Requirements in the Upper Colorado River Basin. *Penn State Department of* State College, PA 2025

Geosciences Graduate Colloquium, Oral Presentation.

Thames, A.B. and Hadjimichael, A.: Assessing Compound Climate Impacts on Agricultural Water Requirements in the Upper Colorado River Basin. *Penn State Interdisciplinary Environmental Research Symposium*, Oral Presentation. — State College, PA 2025

Thames, A.B. and Hadjimichael, A.: Assessing Compound Climate Impacts on Agricultural Water Requirements in the Upper Colorado River Basin. *Penn State Association of Water Students*, Oral Presentation. — State College, PA 2025

Thames, A.B., Hadjimichael, A., and Quinn J.D.: Understanding Compound Climate Impacts to Agriculture Using a Multisite Weather Generator in the Upper Colorado River Basin. *American Geophysical Union*, Poster Presentation. — Washington, D.C. 2024

Climate Intelligence Summer School — Lake Como, Italy
Attendee — 2024

Thames, A.B., Hadjimichael, A., Kukal M.S., and Raj, C.: Assessing the Compound Impacts of Precipitation and Temperature on Agriculture in the Upper Colorado River Basin. *American Geophysical Union*, Poster Presentation. — San Francisco, CA 2023

Thames, A.B. and Foley, B.J.: Producing Feasible Water and Thermal Evolutions for Earth's Mantle Using Monte Carlo Analysis. *American Geophysical Union*, Poster Presentation. — Online 2021

Thames, A.B. and Foley, B.J.: Using Monte Carlo Analysis and Present-Day Constraints on Earth's Water Budget to Produce Feasible Water and Thermal Histories via Reverse-Time Integration. *American Geophysical Union*, Poster Presentation. — Online 2020

Thames, A.B., Brune, W.B., and Miller, D.O.: Global OH Reactivity in the Remote Marine Boundary Layer and the Potential of Missing Reactivity. *American Geophysical Union*, NASA ATom Science Team, Poster Presentation. — Washington, D.C. 2018

Thames, A.B., Brune, W.B., and Miller, D.O.: Research Update #2. *Atmospheric Tomography Mission Science Team Meeting II*, Oral Presentation. — Boulder, CO 2018

Thames, A.B., Brune, W.B., and Miller, D.O.: Research Update #2. *Korea-US Air Quality Mission Science Team Meeting II*, Oral Presentation. — Irvine, CA 2018

Thames, A.B., Brune, W.B., and Miller, D.O.: Measured OH Reactivity in ATom1 and ATom2. *American Meteorological Society*, Poster Presentation. — Austin, TX 2018

Thames, A.B., Brune, W.B., and Miller, D.O.: Research Update #1. *Atmospheric Tomography Mission Science Team Meeting I*, Poster Presentation. — Boulder, CO 2017

Thames, A.B., Brune, W.B., Miller, D.O., and Brosius, A.L.: Research Update #1. *Korea-US Air Quality Mission Science Team Meeting I*, Oral Presentation. — Jeju Island, South Korea 2017

Thames, A.B., Brune, W.B., Miller, D.O., and Brosius, A.L.: Research Update #1. *OH Reactivity Intercomparison Science Team Meeting*, Oral Presentation. — Jülich, Germany 2016

Graduate Teaching Assistant Experience

GEOSC 497: Data Visualization — Spring 2023
The Pennsylvania State University

GEOSC 203: Physical Processes in Geology — Spring 2021, 2022
The Pennsylvania State University

EARTH 002: Earth System and Global Change — Fall 2020
The Pennsylvania State University

EARTH 101: Natural Disasters, Hollywood vs. Reality — Fall 2019
The Pennsylvania State University